

This page was intentionally left blank

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	3 of 30

Change record

Issue	Revision Date	Section/Parag. affected	Reason/Initiation/Documents/Remarks
1	10/09/2026	all	First public release for pipeline version 1.6.11

This page was intentionally left blank

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	5 of 30

Contents

1	Introduction	7
1.1	Scope	7
1.2	What is EDPS?	7
1.3	Main concepts	7
1.4	Installation	8
1.4.1	Prerequisites	8
1.4.2	Installation steps	9
2	Reducing demo data	10
2.1	Setting the workflow	10
2.2	Selecting the input data	11
2.3	Start the reduction	12
2.3.1	Quality plots	13
2.4	Exporting the final products	14
3	The CRIRES+ data reduction flow.	18
3.1	Detector linearity	18
3.2	Dark	19
3.3	Flat	19
3.4	Wavelength calibration	19
3.5	Standard star	20
3.6	Science	20
4	Overview of all the data reduction configuration options	22
4.1	Selection of most appropriate calibrations	22
4.2	Configuration of parameters: the configuration editor	22
5	List of workflow tasks	27
6	Frequently Asked Questions	28

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	6 of 30

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	7 of 30

1 Introduction

1.1 Scope

This document describes how to reduce CRIRES+ data with the `edps-gui` (Graphic User Interface), the dashboard of the ESO Data Processing System (EDPS), which is the recommended interface to reduce data from ESO telescopes. Details on the CRIRES+ data reduction stream and how to configure the reduction to meet specific scientific needs are also given.

For a more extensive documentation on the `edps-gui` itself, consult the dedicated manual [here](#).

For a description of the CRIRES+ pipeline itself, consult the pipeline manual available at: https://www.eso.org/sci/software/pipe_aem_table.html.

Note: this tutorial refers to:

- CRIRES+ instrument pipeline named `cr2res`, version 1.6.11.
- CRIRES+ workflow: `crires.crires_wkf`
- EDPS version 1.5.7.
- `edps-gui` version 0.91.

1.2 What is EDPS?

The ESO Data Processing System (EDPS) is a framework to run ESO's data processing pipelines and it is meant to eventually replace the previous [ESOReflex environment](#). The general principles of EDPS have been described by [Freudling, Zampieri, Coccato et al. \[2024, A&A, 681, A93\]](#). Please refer to that paper if you have used EDPS for research resulting in a scientific publication.

Each of ESO's data processing pipeline consists of a series of standalone programs called *recipes*. Each recipe is designed to process certain type(s) of input data. The processing of these input data typically requires a range of auxiliary files such as calibration files. EDPS is designed to select appropriate input data for the different recipes of a pipeline, and execute them in sequence. This is done by specifying for each pipeline the workflow for organizing data and executing the recipes. Then, the workflow can be used to process a set of data fully automatically.

1.3 Main concepts

EDPS is an environment designed to execute the recipes of an instrument pipeline according to a series of instructions. The main concepts in EDPS are:

- **Workflow and reduction cascades.** A workflow is a series of instructions designed to reduce data with an instrument pipeline in potentially multiple ways, by carrying on a sequence of tasks. Each workflow can define multiple reduction cascades, depending on the scientific needs. For example, the same workflow

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	8 of 30

can be used to process data following different strategies that trigger different reduction steps (e.g. in one strategy flux calibration can be omitted) or different end-points (e.g., combine different science exposures, or stop after the reduction of individual exposures without combining them). Each of these "strategies" defines a "reduction cascade".

- **Task, jobs, and recipes.** A task is an element in the workflow that performs a given step of the data reduction cascade. Tasks are often associated to a recipe of the underlying instrument pipeline. A job is a work unit in a processing environment, that runs a recipe on a set of input data with a set of recipe parameters. A single task can generate several jobs: for example, a "bias" task, can generate multiple jobs, each of the running the bias recipe on a different set of input files.
- **Dataset.** A dataset is a collection of files, that are needed to perform the data reduction as specified by the workflow. It consists, for example, of one or more science files plus the calibrations needed to process them. In EDPS, datasets have a hierarchical structure, which highlights the connections between the various files and tasks (e.g., task A is an input to task B).
- **Target and Target category.** The "target", or the "target task" is the end point of the reduction cascade. When specifying a target, EDPS will process all and only the files needed to execute it. For example, if my target is "science", and the science files need the bias files, EDPS will process only the biases that have been selected to process those science files; then it processes the science using the product of the bias reduction. However, if my target is bias, then EDPS will process all and only the bias files, regardless they are not used by any science. In this case, EDPS does not process the science, as it has already reached the end reduction point (e.g., process all biases). The "Target category" is a group of targets that have similar purposes. For example, the target category "science", includes all the tasks that deliver final scientific products, the target category "qc1calib" includes all and only the tasks that processes calibrations (e.g., bias, flat fields, standard stars).

1.4 Installation

1.4.1 Prerequisites

Prerequisites for a well functioning installation of EDPS and EDPS-gui are:

- Recent Firefox or Chrome browser, Python 3.11 or higher (but there are issues with Python 3.14).
- At least one ESO pipeline with EDPS workflow should be in your system. To install the desired ESO pipelines, follow the instructions in the ESO pipelines pages. NOTE: the `apptainer` installation method is currently not supported. After the installation, the `esorex` command must be in the path. To test whether the installation was successful, type

```
esorex --recipes
```

A list of available recipes should appear.

- Install `graphviz`, `fv`, and `ds9`, which have to be included in the system path (defining aliases not enough). On linux, `Graphviz` can be easily installed via:

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	9 of 30

```
sudo apt install graphviz (Debian, Ubuntu)
sudo dnf install graphviz (Fedora)
```

Check the [Graphviz webpage](#) for installation instructions for other OS.

`fv` and `ds9`, are optional. To install them, follow the instructions in corresponding webpages. You can test whether these three packages are installed and their paths are correctly set by typing on a terminal:

```
dot -V
fv -version
ds9 -version
```

1.4.2 Installation steps

To install EDPS follow these steps:

- Create a new Python virtual environment and activate it:

```
python3 -m venv edpsgui
. edpsgui/bin/activate
```

Make sure the `python3` version is 3.11 or higher, but not 3.14.

- Install the required packages:

```
pip install --extra-index-url \
https://ftp.eso.org/pub/dfs/pipelines/repositories/stable/src \
edps edpsgui edpsplot adari_core
```

To run the `edps-gui` type from a terminal (with the active environment):

```
edps-gui
```

Important note. The first time `edps-gui` is executed, you will be asked to specify the directory where the reduction products (fits files and quality plots) will be stored. The default location is `$HOME/EDPS_data`. During the first execution, a configuration file named `application.properties` will also be saved in the directory (newly created) `$HOME/.edps`.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	10 of 30

2 Reducing demo data

Follow this procedure to quickly reduce CRIRES+ demo data. We assume that the EDPS, `edps-gui`, the CRIRES+ pipeline and its associated demo data are installed in your system. For general instructions on how to install EDPS and the pipeline, see Section 1.4 or please visit: https://www.eso.org/sci/software/pipe_aem_main.html.

2.1 Setting the workflow

Proceed as follows:

1. If not done already, activate the EDPS virtual environment, defined during installation (Sect. 1.4).
2. Start the `edps-gui` dashboard by typing:

```
edps-gui
```

The `edps-gui` dashboard will start in a browser window (Figure 1).

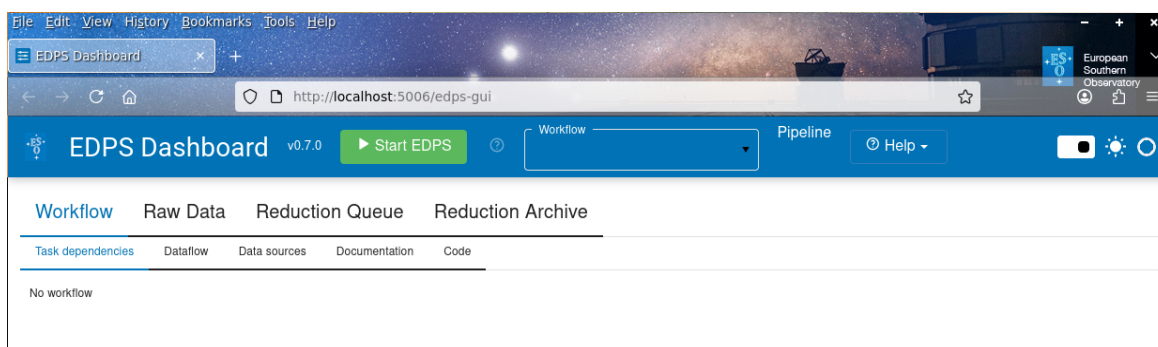


Figure 1: The empty `edps-gui` Dashboard; the underlying EDPS engine has not yet been started and no workflow has been loaded.

3. Optionally, before starting EDPS, one can specify new settings by pressing Help → Settings (Figure 2).

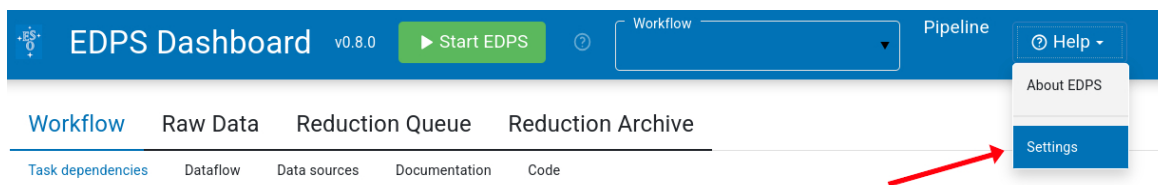


Figure 2: The “Help” → “Settings” menu.

4. On the browser window with the dashboard, press the button ‘Start EDPS’.
5. Choose the `crires.crires_wkf` workflow from the list in the ‘Workflow’ field. The workflows offered in this selector depend on the installed pipelines. The graphic workflow representation will appear as in Figure 4.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	11 of 30

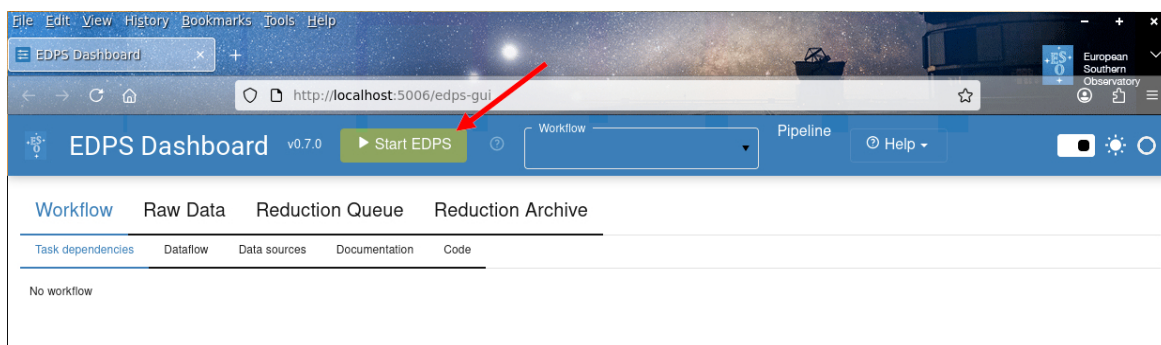


Figure 3: The “Start EDPS” button.

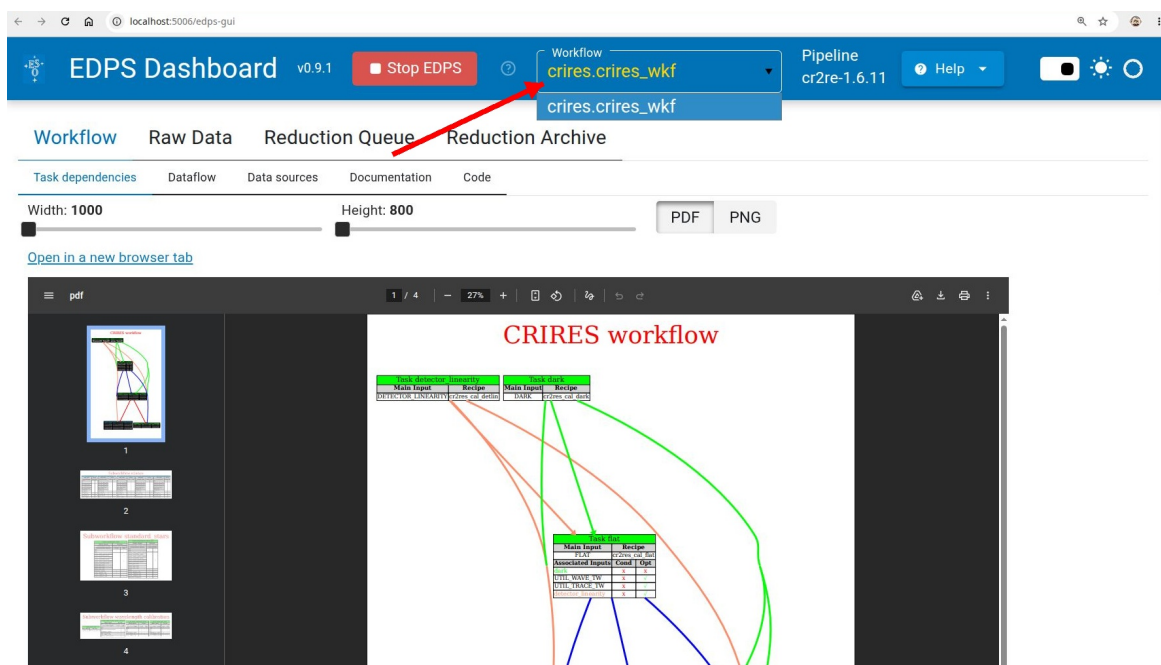


Figure 4: The edps-gui with the CRIRES+ workflow loaded.

2.2 Selecting the input data

1. Press ‘Raw Data’ to enter the corresponding tab, as in Figure 5.

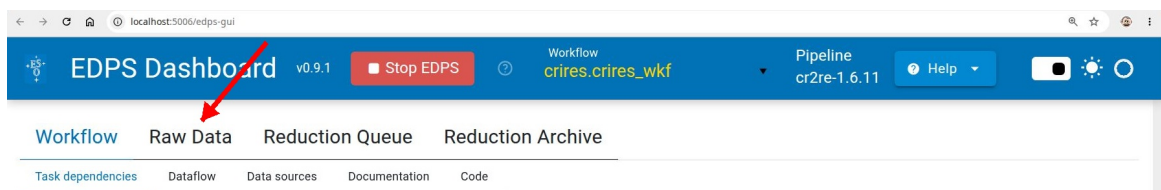


Figure 5: How to select RAW data Tab.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	12 of 30

2. Press ‘Select Inputs’. A selection window will appear that allows to select data that are stored on a local disk (Figure 6).

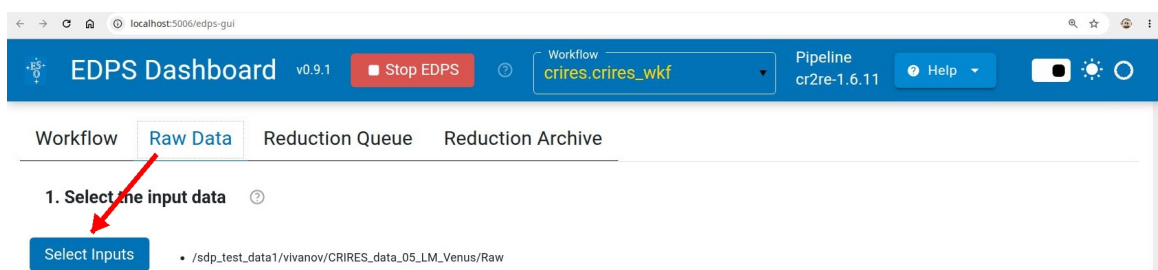


Figure 6: How to select input data.

3. (Optional). Select the reduction target, configure the workflow parameter and specify the association preferences. These steps are optional. For more information see Section 4.
4. Press ‘Create Datasets’. A list of datasets appears, one line for each set of science data (Figure 7).

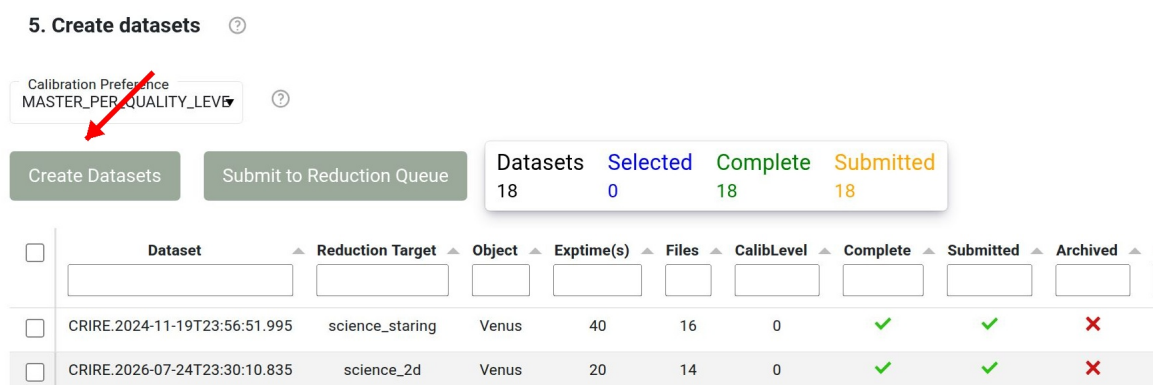


Figure 7: How to inspect the input data directory to create datasets.

5. Choose the datasets that should be processed (Figure 8) and send them to the data reduction queue by pressing ‘Submit to Reduction Queue’. Note that this action does not start the reduction automatically.

2.3 Start the reduction

1. Press the ‘Reduction Queue’ tab (Figure 9).
2. Select the datasets you’d like to reduce.
3. (Optional). Configure the workflow and recipe parameters by pressing the wheel button  to open the configuration editor. See Section 4.2 for more information on the configuration editor.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	13 of 30

5. Create datasets

Calibration Preference: RAW_PER_QUALITY_LEVEL

Create Datasets Submit to Reduction Queue

Datasets: 18 Selected: 3 Complete: 18 Submitted: 0

Dataset	Reduction Target	Object	Exptime(s)	Files	CalibLevel	Complete	Submitted	Archived	det.ndit	det.read.curname	det.seq1.dit	dpr.type	ins.sli
☒ CRIRE.2026-07-23T23:30:54.761	science_2d	Venus	20	14	0	☒	☒	☒	3	New_RR_UpTheRamp	20	OBJECT	w...
☐ CRIRE.2026-07-23T23:45:03.084	science_2d	Venus	20	14	0	☒	☒	☒	3	New_RR_UpTheRamp	20	OBJECT	w...
☒ CRIRE.2026-07-24T23:12:40.962	science_2d	Venus	20	14	0	☒	☒	☒	3	New_RR_UpTheRamp	20	OBJECT	w...
☐ CRIRE.2026-07-23T23:51:23.076	science_2d	Venus	20	15	0	☒	☒	☒	3	New_RR_UpTheRamp	20	OBJECT	w...
☒ CRIRE.2026-07-24T23:19:13.195	science_2d	Venus	20	14	0	☒	☒	☒	3	New_RR_UpTheRamp	20	OBJECT	w...

Figure 8: How to send the selected datasets to the Reduction Queue for processing.

EDPS Dashboard v0.9.1 Stop EDPS Workflow: crises.crires_wkf Pipeline: cr2re-1.6.11 Help

Workflow Raw Data Reduction Queue Reduction Archive

1. Select the input data

Select Inputs

- /diskb/vivanov/CRIRES_data_05_LM_Venus_Cons/Raw
- /diskb/vivanov/CRIRES_data_05_LM_Venus

Figure 9: How to select Reduction Queue tab.

4. Press the 'Reduce' button (Figure 10). The selected data will now be processed with the configured parameters.

EDPS Dashboard v0.9.1 Stop EDPS Workflow: crises.crires_wkf Pipeline: cr2re-1.6.11 Help

Workflow Raw Data Reduction Queue Reduction Archive

All workflows Selected workflow

Reduce Archive Delete

Reductions: 3 Selected: 3 New: 3 Pending: 0 Running: 0 Completed: 0 Aborted: 0 Failed: 0

Dataset	Reduction Target	Object	Configuration Date	Status	Comment	Parameters
☒ CRIRE.2026-07-23T23:30:54.761	science_2d	Venus	2026-09-09T15:21:53.405	NEW		

Figure 10: Reduce.

Congratulations! You reduced your first data with the EDPS dashboard! All the reduced data are saved in the EDPS_data directory specified when executing edps-gui for the first time.

2.3.1 Quality plots

It is possible to inspect the information on each job, such as quality plots showing the products (for the most important jobs and products), the list of inputs and output files, the recipe parameters and logs. All the information

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	14 of 30

on the job processing can be inspected from the ‘Reduction Queue’ window. While quality plots are produced only for completed jobs, the other relevant information is available also for failed jobs.

The information associated for the main product can be inspected by pressing the magnifying glass symbol at the right side of each dataset. To inspect the information associated to other individual jobs (e.g., calibrations), proceed as follows:

- Expand the desired dataset by pressing the black arrow on its left. The list of jobs will appear with the associated status (COMPLETED, RUNNING, PENDING, MISSING, ABORTED, FAILED)
- Press the magnifying glass symbol at the right side of the job you want to inspect.

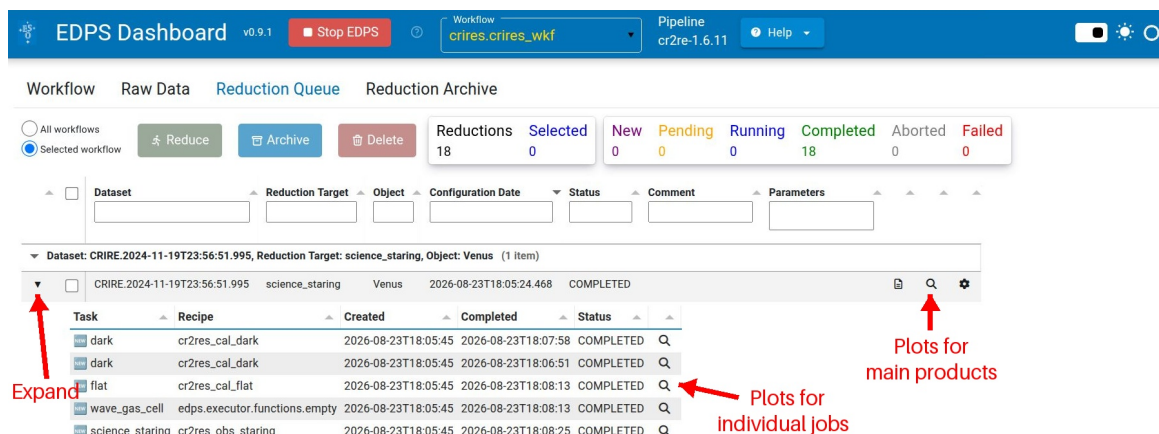


Figure 11: How to look for job information from the Reduction Queue tab.

2.4 Exporting the final products

Completed reductions can be ‘Archived’ (i.e. declared ‘completed’ because no more work is needed) and removed from the Reduction Queue. Additionally, even if all products for all tasks are saved in the EDPS_data directory, the most important products can be ‘exported’ to a desired location.

To do so, proceed as follows:

1. In the ‘Reduction Queue’ tab, select the dataset and the dataset for which you want to export the final products, and press the ‘Archive’ button.
2. Go in the Reduction Archive tab and click on the ‘Export’ button. A new tab window appears where you can indicate the directory you want to copy your final products; finally press "Export" to copy the data.

Exported products are organized by ‘DATASET’ (named as the first scientific exposure of the dataset), and ‘TIMESTAMP’ (time of start of reduction)

The more important final products saved in the individual directories are:

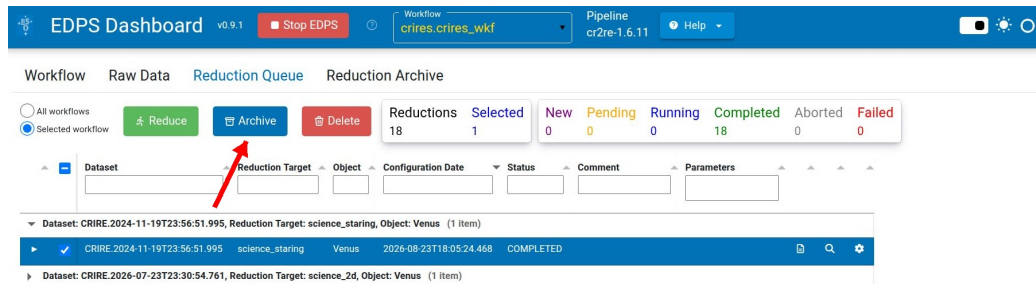


Figure 12: How to archive a completed reduction from the Reduction Queue tab.

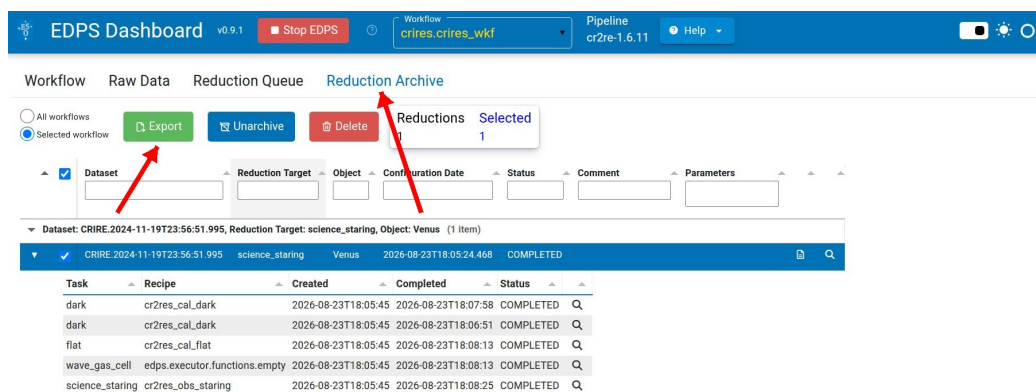


Figure 13: The reduction archive tab. This table contains all the different configurations of datasets that are declared "finished" and removed from the Reduction Queue. From this page, the user can export the most important files into a desired local directory.

- Darks:
 - master dark (`PRO.CATG = CAL_DARK_MASTER`)
 - bad pixel mask (`PRO.CATG = CAL_DARK_BPM`)
- Flat:
 - master flat (`PRO.CATG = CAL_FLAT_MASTER`)
 - blaze function (`PRO.CATG = CAL_FLAT_EXTRACT_1D`)
 - bad pixel mask (`PRO.CATG = CAL_FLAT_BPM`)
 - slit function (`PRO.CATG = CAL_FLAT_SLIT_FUNC`)
 - 2D slit model (`PRO.CATG = CAL_FLAT_SLIT_MODEL`)
 - tracewave table with order tracing (`PRO.CATG = CAL_FLAT_TW`)
 - tracewave merged table (`PRO.CATG = CAL_FLAT_TW_MERGED`)
- Wavelength:
 - tracewave table with order tracing and wavelength calibration (`PRO.CATG = CAL_FLAT_TW`)
 - 2D wavelength map (`PRO.CATG = CAL_WAVE_MAP`)

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	16 of 30

×

Selected reductions

Dataset ▲	Configuration Date ▲	Status ▲	Parameters ▲
▼ Dataset: CRIRE.2024-11-19T23:56:51.995 (1 item)			
CRIRE.2024-11-19T23:56:51.995	2026-08-23T18:05:24.468	COMPLETED	

☒ Export all selected reductions
☐ Export the latest reduction for each selected dataset

Output directory

Export

Close

Figure 14: The EXPORT dialogue window, where the user can decide which reduced configuration to save and where.

- table with extracted multi-order 1D arc FP spectrum ($PRO.CATG = CAL_WAVE_EXTRACT_1D$)
- fits table with diagnostics for individual detected lines ($PRO.CATG = CAL_WAVE_LINES_DIAGNOSTICS$)
- Science Nodding:
 - combined 2D spectra for nodding positions A and B separately ($PRO.CATG = OBS_NODDING_COMBINEDA/B$)
 - table with extracted multi-order 1D spectrum for the two nodding positions separately, and for them combined together ($PRO.CATG = OBS_NODDING_EXTRACTA/B/_COMB$)
 - model 2D spectrum for nodding positions A and B separately ($PRO.CATG = OBS_NODDING_SLITMODELA/B$)
 - slit function (slit profile) for nodding positions A and B separately ($PRO.CATG = OBS_NODDING_SLITFUNCA/B$)
 - extracted 1D IDP products (combining all orders) for the two nodding positions separately, and for them combined together ($PRO.CATG = OBS_NODDING_EXTRACTA/B/C_IDP$)
 - tracewave tables for nodding positions A and B separately ($PRO.CATG = OBS_NODDING_TWA/B$)
- Science Staring:

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	17 of 30

- combined 2D spectrum (`PRO.CATG = OBS_STARING_COMBINED`)
- table with extracted multi-order 1D spectrum (`PRO.CATG = OBS_STARING_EXTRACT`)
- model 2D spectrum (`PRO.CATG = OBS_STARING_SLITMODEL`)
- slit function (slit profile) (`PRO.CATG = OBS_STARING_SLITFUNC`)
- extracted 1D IDP product (combining all orders) (`PRO.CATG = OBS_STARING_IDP`)
- Science 2D (offset mode observations):
 - calibrated 2D spectrum (`PRO.CATG = OBS_2D_CALIBRATED`)
 - table with extracted multi-order 1D spectrum (`PRO.CATG = OBS_2D_EXTRACT`)

For a full list of the products see Sec. 11.2 of the CRIRES+ Pipeline Manual.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	18 of 30

3 The CRIRES+ data reduction flow.

The overall data flow of the CRIRES+ pipeline is displayed in Figure 15.

The reduction cascade is organized in tasks, which represent well-defined steps in the process. Tasks can be grouped inside sub-workflows. Each task runs a recipe; the detailed description of the algorithms, input, outputs and recipe parameters used in each recipe are available in the pipeline manual. Here, we present only the description of most important features.

The `crires.crires_wkf` EDPS workflow is designed to execute the tasks that deliver the final reduced data cube for each dataset. It can be either the product of a single exposure, or the combination of multiple exposures. Only calibrations needed by the selected the scientific exposures are processed.

It is possible to set EDPS to perform the data reduction until a certain step of the reduction chain (e.g. to reduce only standar stars, or only flat fields). This is done by specifying the desired tasks in the field **Select reduction target** of the **Raw Data** tab.

The reduction steps of the `crires.crires_wkf` workflow are listed below. Before starting the reduction,

the parameters of the recipes associated to each task can be configured by pressing the button  close to each dataset configuration. See for more info on the configuration editor [4.2](#)

The CRIRES+ data reductions is closely intertwined with the observing mode. Typically, the CRIRES+ observations are carried out in the so-called nodding mode, in which the telescope is “nodding” so the target is placed at two positions along the slit – nodding positions A and B. In this case, in sequential frames the instrument sees sky at the position where the target was located in the preceding or the following exposure. This allows to remove the sky contribution to zeroth order just by taking the difference between frames taken at two different nodding positions. Brighter targets that require shorter exposures, thereby minimizing the sky emission, may be observed in stare mode – in this case the telescope remains stationary, so the target is always kept on the same position on the slit. In this case, if there is any significant sky emission, it must be extracted from off-target apertures, and then subtracted from the target aperture. Finally, large targets that occupy more than half of the slit are typically observed in offset mode – the telescope alternates between target exposures and clear sky exposures. Applying small offsets along the slit, called jittering, may be introduced in all these cases. Jitters help to minimize the effect of poor detector cosmetics, because the same point of the target is observed with different pixels in different exposures. For further details, see CRIRES+ Pipeline Manual.

3.1 Detector linearity

The detector’s response is non-linear. In other words the measured ADUs are not linearly proportional to the number of incident photons and the correction is different for each pixel. The correction is applied on a calibration data product (PRO.CATG=CAL_DET LIN_COEFFS) provided by the ESO Archive. The correction is applied by the `cr2res_cal_detlin` recipe. At $\sim 30,000$ ADU, the non-linearity is at $\sim 10\%$ level, and the correction is accurate to 0.5% . During the derivation of the non-linearity correction bad pixels that do not follow the expected linearity behavior are identified and masked out. Further details are available in the CRIRES+ Pipeline manual, Sec. 6.2.5.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	19 of 30

3.2 Dark

Infrared detectors are sensitive to the thermal electrons existing inside the detector itself, and despite the cooling, they always exhibit some intrinsic signal that is proportional to the integration's length. The dark frames can show structures because of the different characteristics of the ADCs and heat sources inside the instrument (e.g., amplifier glow). To remove the dark signal and associated structures, a master dark is subtracted from each frame. It is true that in some cases the dark can be subtracted together with the sky frames – for example in nodding mode observations. However, subtracting the dark in advance is often preferable, because it improves the statistic calculations necessary to perform rejections by removing the patterns discussed above. The master dark is generated by the `cr2res_cal_dark` recipe and is subtracted in various science recipes using the utility recipe `cr2res_util_calib`. The dark signal in the flats is removed, typically by subtracting the lamp OFF frames from the lamp ON frames. Note that removing the dark signal does not remove the noise contribution that comes from it. The dark frames are used to identify a type of bad pixels called “hot” – that always exhibit high values. Further information is available in the CRIRES+ Pipeline manual, Sec. 4.4.

3.3 Flat

Even the best detectors are not uniform, and individual pixels show slightly different sensitivity to light. The usual approach to correct for this effect is to observe a uniformly illuminated source and derive the true sensitivity of each pixel. Note that the pixels may suffer from intra-pixel sensitivity as well, which cannot be corrected by means of flat fielding. The master flat is generated by the `cr2res_cal_flat` and it is applied by various science recipes using the utility recipe `cr2res_util_calib`. The illuminated flat frames are used to identify a type of bad pixels called “dead” – that always exhibit low, near-zero values. This step is also used to identify the location of the spectral orders. The result is stored in a tracewave table with geometric information about the orders. Further information is available in the CRIRES+ Pipeline manual, Sec. 4.4.

3.4 Wavelength calibration

The CRIRES+ wavelength calibration depends on the mode, particularly on the observed wavelength range. The user can control it with the `--wl_method` parameter of the `cr2res_util_wave` recipe. Generally, the calibration is derived by cross-correlating the extracted lamp spectrum and a synthetic Uranium–Neon spectrum. For the Fabry–Pérot, the wavelength calibration is derived with a fit to the 2D spectrum, but this method still needs the UNe lamp calibration as an initial guess, because the Fabry–Pérot spectra show regular, repeatable patterns. The individual calibration strategies are implemented in the `cr2res_util_wave` recipe, and their combination is implemented in `cr2res_cal_wave` recipe. These recipes update the tracewave table derived in the previous step with the wavelength information. Further information is available in the CRIRES+ Pipeline manual, Sec. 6.2.3. Note that for gas cell wavelength calibration (that only works for the shorter wavelengths) a special external tool¹ is available for improving the accuracy, and it has been demonstrated to improve the precision down to 2-3 m s⁻¹.

¹VIPER (Velocity and IP Estimator): https://mzechmeister.github.io/viper_RV_pipeline/

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	20 of 30

3.5 Standard star

The instrument response is derived by observing a set of flux-calibrated standard stars. Their true spectra are stored in a FITS table `std_photom.fits` that has self-explanatory column names and is located in the calibration sub-directory of the directory where the pipeline is installed: `.../install/calib/cr2re-1.6.11/`. The pipeline derives the response function by comparing the shape of the observed spectrum with the shape of the supplied flux calibrated spectrum in that table.

An example of full manual command-line sequence how to calibrate the CRIRES+ data is available in the CRIRES+ Pipeline Manual, Sec. 5.5. Inspecting it can be helpful for users, because EDPS follows the same strategy, even though it is executed automatically.

3.6 Science

This step is observing mode and wavelength dependent, and it is executed by one of the following recipes: `cr2res_obs_staring`, `cr2res_obs_nodding`, `cr2res_obs_2d`, and `cr2res_obs_pol`. These recipes apply all calibrations listed above and (except `cr2res_obs_2d`) perform extraction of science targets. Both regular aperture and optimal (recommended) extraction methods are available. That step is the most likely to need refinement and the pipeline users will benefit from experimenting with the extraction parameters. These include:

- `--extract_height` that defines the extraction height from the edge-tracing polynomials, defined in the tracewave file; this is particularly important if there are multiple sources on the slit or if the target is not well centered.
- `--extract_swath_width` that defines the length of the regions in which, among other things, the target profile for the optimal extraction is defined; extracting spectra of fainter targets benefits from higher values of this parameters.
- `--extract_oversample`, which controls the oversampling of the target profile; experiments suggest that values >12 work well.

Detailed information about the optimal extraction can be found in the CRIRES+ Pipeline Manual, Sec. 6.1.1. Note that the optimal extraction does not work for targets with variable spatial profiles, for example resolved galaxies and disks that may have different profiles for the continuum and for various emission lines.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	21 of 30

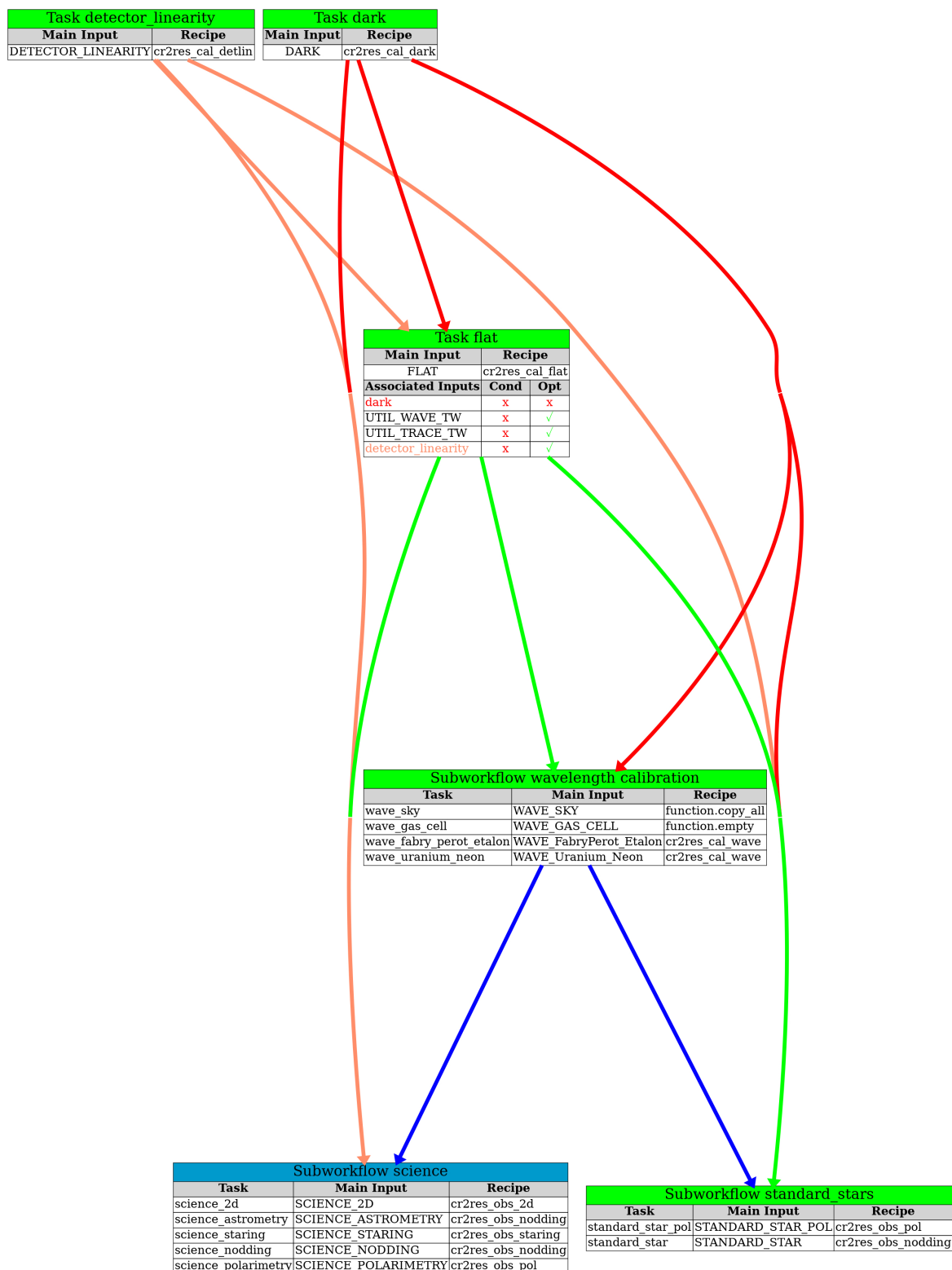


Figure 15: The data reduction cascade of the CRIRES+ workflow.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	22 of 30

4 Overview of all the data reduction configuration options

4.1 Selection of most appropriate calibrations

By default, EDPS associates raw calibrations to the reduction process. It is also possible to use pre-processed calibrations (a.k.a. master calibrations) if available, in order to speed up the reduction. The preference can be specified in the Raw Data tab, before creating the datasets.

Possible values of the Calibration Preferences are:

- **raw_per_quality_level**: At equal quality of reduction, association of raw calibrations is preferred. This is the default.
- **master_per_quality_level**: At equal quality of reduction, association of master calibrations is preferred.
- **raw**: Association of raw calibration is preferred, despite the quality of results.
- **master**: Association of master calibration is preferred, despite the quality of results.

When master calibrations are used, the reduction step needed to process raw calibrations are not executed. The reduction then moves directly to the process of scientific exposures.

For example, if reduction speed for a quick check is preferred over a high quality reduction, one can select "master". In this case, old master calibrations are associated even if there are raw calibrations closer in time (and therefore more likely to ensure better quality products).

The quality level that the selected calibrations deliver is indicated close to each dataset in the 'Raw input' tab, under the column 'CalibLevel'. CalibLevel=0 indicates that calibrations that follow the rules of the instrument calibration plans have been selected. The higher the number, the poorer the quality of the products.

4.2 Configuration of parameters: the configuration editor

The data reduction of each dataset can be configured according to the scientific needs using an appropriate configuration editor.

The EDPS workflows contain two types of parameters and they both have default values that can be modified to improve the data reduction.

- **Workflow parameters** (for some workflows only) are global and they are applied to the entire workflow. They are accessible both in the 'Raw Data' tab, prior to the creation of a dataset, and in the 'Reduction Configuration' editor, in the 'Reduction queue' tab. Note: some workflow parameters were already configured before creating the dataset and sending it to the reduction queue. Here, they can be changed again. Please, note that the parameters have an effect only on the files that are already in the dataset. If one specifies a parameter that should include extra files in the dataset (e.g., the inclusion of more calibrations), files are not added and the reduction might fail. If you need to change a parameter that modifies the dataset content, please go back to the Raw data tab and create a new dataset.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	23 of 30

- **Recipe parameters** are specific to the individual recipes and can be configured per task. They are accessible in the ‘Reduction Configuration’ editor, in the ‘Reduction queue’ tab.

This editor allows to configure the data reduction for a given dataset by specifying workflow and recipe parameters.

Note: some workflow parameters were already configured before creating the dataset and sending it to the reduction queue. Here, they can be changed again. Please, note that the parameters have an effect only on the files that are already in the dataset. If one specifies a parameter that should include extra files in the dataset (e.g., the inclusion of more calibrations), files are not added and the reduction might fail. If you need to change a parameter that modifies the dataset content, please go back to the Raw data tab and create a new dataset.

To open the editor, click on the wheel button  next to the dataset you desire to configure the reduction for. A window with the configuration editor appears as shown Figure 16.

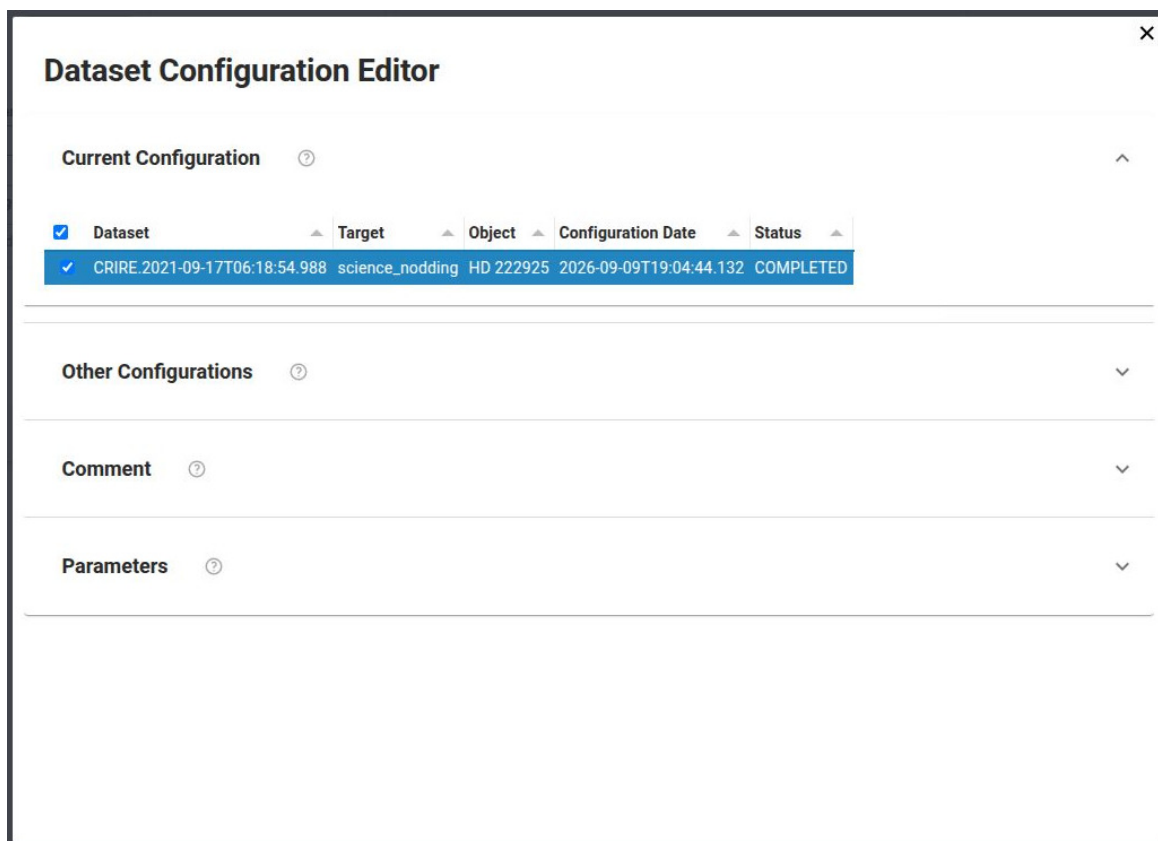


Figure 16: The Reduction Configuration editor. It contains 4 sections, that indicate the current configuration, list of other configurations to set, comments to insert, and the parameters to modify.

The editor is divided into 4 parts, which can be accessed pressing the corresponding expansion arrow.

- **Current configuration.** It indicates the name of the selected configuration for a given dataset (Figure 17).

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	24 of 30

- **Other configurations.** It allows to specify other configurations, to which the changes shall be copied to (Figure 18).
- **Comment.** It allows to specify a comment to describe the configuration. It is possible to append or replace a comment (Figure 19). Comments can be changed on all configurations. It is possible to save the comment for the current configuration only, or for all the selected configurations.
- **Parameters.** A window as in Figure 20 appears.

The window allows to:

- Select the parameter set. A pre-determined list of workflow parameters and recipe parameters for a given use case. For the majority of the cases, the "science" parameter set can be used.
- Edit the workflow parameters. These are parameters that regulates the reduction strategy, e.g. whether to use a given calibration or not, or to trigger a certain reduction step. Note that if the changes imply that some files not in the dataset are needed, the reduction might fail. In case, go back to the raw data tab, edit the workflow parameters there, and recreate the datasets.
- Edit the recipe parameters. These are parameters associated to the recipe of a given task. Note: the same recipe parameters can be configured differently for the tasks that run the same recipe. Default parameters are shown (albeit some parameters can be dynamic, e.g. 'EDPS' changes their value depending on the type of input data).

Change the values according to the needs and then select whether to save it to the current or the selected configurations. Note, complete configurations cannot be modified, new configurations will be automatically created instead.

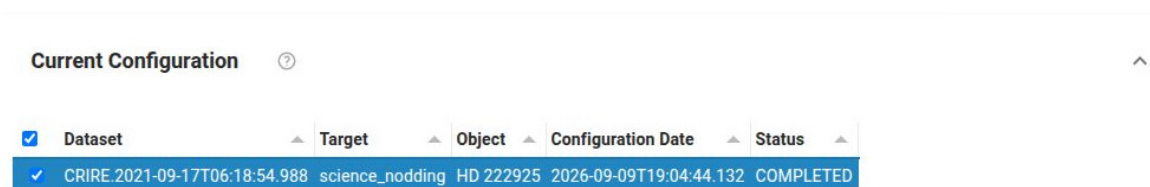


Figure 17: The first part of the Reduction Configuration Editor, that indicates the selected configuration.

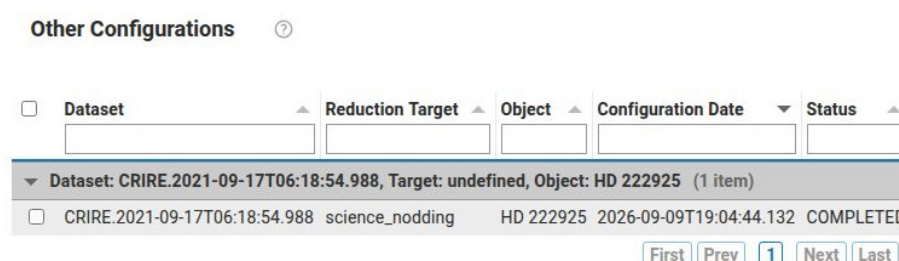


Figure 18: The second part of the Reduction Configuration Editor, that indicates other configurations for which we'd like to apply the changes.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	25 of 30

Comment ? ^

Comment

This is a comment describing the reduction

Save

Copy to selected configurations

☒ append
☐ replace ?

Figure 19: The third part of the Reduction Configuration Editor, that allows to specify a comment to the selected configurations.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	26 of 30

Parameters ?

Parameter set

science_parameters

Workflow parameters

Parameter	Default value	Custom value
\$combination_kwd	tpl.start	

Click on a parameter to view its description

Recipe parameters

Task

dark

Parameter	Default value	Custom value
cr2res.cr2res_cal_dark.detector	0	
cr2res.cr2res_cal_dark.bpm_method	DEFAULT	
cr2res_cal_dark.bpm_kappa	-1.0	
cr2res_cal_dark.bpm_lines_ratio	0.5	
cr2res_cal_dark.ron_hsize	6	
cr2res_cal_dark.ron_nsamples	100	
cr2res_cal_dark.gain	2.1	
cr2res_cal_dark.collapse.method	MEAN	
cr2res_cal_dark.collapse.sigclip.kappa-low	3.0	
cr2res_cal_dark.collapse.sigclip.kappa-high	3.0	
cr2res_cal_dark.collapse.sigclip.niter	5	
cr2res_cal_dark.collapse.minmax.nlow	1.0	
cr2res_cal_dark.collapse.minmax.nhigh	1.0	
cr2res_cal_dark.collapse.mode.histo-min	10.0	
cr2res_cal_dark.collapse.mode.histo-max	1.0	
cr2res_cal_dark.collapse.mode.bin-size	0.0	
cr2res_cal_dark.collapse.mode.method	MEDIAN	
cr2res_cal_dark.collapse.mode.error-niter	0	

Save

Save as new configuration

Copy to selected configurations

?

Figure 20: The fourth part of the Reduction Configuration Editor, that allows to specify the parameters sets and the recipe parameter per task. These settings can be applied to the "Selected Configuration" (Fig. 17) or to the "Other Configurations" (Fig. 18).

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	27 of 30

5 List of workflow tasks

This is the list of all the tasks and associated recipes in the CRIRES+ workflow. Only some of them are needed for scientific reduction, they are indicated by the flag "yes" (triggered by default) or "optional" (triggered only if requested by a workflow parameter). Other tasks are not used for scientific reduction (they are indicated by the flag "no"), they are mainly used for instrument monitoring and they can be executed only by specifying them as target. Note that, when a task is specified as target, all the tasks that generate the calibrations needed for it are automatically executed.

TASK	RECIPE	Used in science reduction	Notes
detector_linearity	cr2res_cal_detlin	optional	
dark	cr2res_cal_dark	yes	
flat	cr2res_cal_flat	yes	
wave_uranium_neon	cr2res_cal_wave	yes	
wave_gas_cell	cr2res_cal_wave	yes	
wave_sky	cr2res_cal_wave	optional	
wave_fabri_perot_etalon	cr2res_cal_wave	yes	
standard_star	cr2res_util_calib	yes	
standard_star_pol	cr2res_util_calib	yes	
science_staring	cr2res_obs_staring	yes	
science_nodding	cr2res_obs_nodding	yes	
science_2d	cr2res_obs_2d	yes	
science_polarimetry	cr2res_obs_pol	yes	
science_astrometry	cr2res_obs_nodding	yes	

Table 5.0.0: CRIRES+ pipeline tasks overview

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	28 of 30

6 Frequently Asked Questions

- **Q1) Where can I find the final reduced data?**

Answer: all the products of all the datasets and the reductions are saved into the EDPS_data directory, specified when executing the edps-gui for the first time. One can decide to export only the final products for selected datasets and only for the desired reduction attempts into another location for further analysis. See Section 2.4 for further instructions.

- **Q2) How do I stop the application?**

Answer: Proceed as follows:

1. Press “Stop EDPS” in the Dashboard.
2. Type Ctrl-C in the terminal where the application is running. If the application doesn’t terminate, type Ctrl-C again.
3. Alternatively, kill the ‘panel serve’ process on your system, for example:

```
ps -e | grep panel # get the process ID of the gui (<pid>).
kill -9 <pid>
```

- **Q3) I have closed the browser window where the application is running. How can I reopen the application?**

Answer: Point your browser to: `http://localhost:5006/edps-gui`

- **Q4) Where can I find some data that I can use to test the application?**

Answer: Install the ‘datademo’ package provided with the pipeline installation or download the “Demo Data” package from https://www.eso.org/sci/software/pipe_aem_table.html.

Please note that the demo data can be large (tens of Gigabytes).

A convenient script to download demo data for any pipeline is also available and can be used from the command line:

```
curl -O https://eso.org/sci/software/apptainer/eso_download_demodata.sh
bash ./eso_download_demodata.sh
```

- **Q5) How can I start the edps-gui if the following message appears?**

```
Cannot start Bokeh server, port 5006 is already in use
```

Answer: The panel server was not closed properly. Kill it by typing:

```
ps -e | grep panel # get the process ID of the gui (<pid>).
kill -9 <pid>
```

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	29 of 30

• **Q6) How do I get additional support on EDPS or data reduction in general?**

Answer: For suggestions, questions, or feedback in general, please open a ticket with the EDPS Support team. This https://support.eso.org/new-ticket?ticket%5Bticket_field_13%5D%5Bdata%5D=227 should take you directly to a webpage for creating and EDPS feedback ticket, but in case you want to navigate there 'manually', go to <https://support.eso.org>, login, click on "Submit Helpdesk Ticket", and specify the Help topic: "Post Observations", "ESO Data Processing System [EDPS]".

• **Q7) I have a lot of disk space, but when I install EDPS with pip or an ESO pipeline with Homebrew I get the error message: Cannot mkdir: No space left on device. How do I fix it?**

Answer: This depends on how much disk space is allocated to the /home, /var, and /tmp directories. The final solution would be to resize the space allocated to the in the organization of the filesystem. However, we list here few tricks that might do the job.

- Clearing the pip .cache to make space for new packages. Type the command:

```
pip cache purge
```

before installing EDPS.

- Redirect the cache, Homebrew temporary build directories into a partition with enough space. Set some of the following environmental variables in your .bashrc file:

```
export HOMEBREW_CACHE=<path_to_new_cache_directory>
export XDG_CACHE_HOME=<path_to_new_cache_directory>
export HOMEBREW_TEMP=<path_to_new_temporary_directory>
export TMPDIR=<path_to_new_temporary_directory>
```

The first moves only the location of Homebrew cache, the second the cache of most applications (instead of the default /home/username/.cache), the third moves the directory where Homebrew builds, extracts, and saves temporary files (instead of the defaults /tmp and /var/tmp). The last changes the global system temporary directory and affects most of the linux commands.

- As extreme measure, one can move the /home/linuxbrew/.linuxbrew directory somewhere else, and create a symbolic link in /home/linuxbrew. For example:

```
cd /home/linuxbrew
mv -f .linuxbrew <path_to_new_directory>
ln -s <path_to_new_directory> .linuxbrew
```

Important note: this operation might break some internal links. Recipes requiring external packages such as telluriccorr might not work (impacts on KMOS, XSHOOTER, FORS2, and MOLECFIT pipelines).

• **Q8) How can I manually run the recipes executed by EDPS?**

Answer: The directory ~/EDPS_data/CRIRES/ contains individual subdirectories for each recipe, e.g., dark, flat, wave_gas_cell, science_2d, and science_staring – one for each recipe. They contain randomly named sub-directories for each execution of the particular recipe, e.g., 0ea47c20-26dd-4ffd-8de2-9cad4bbf515a, 2a73f7e3-aa4f-4853-baff-33de29c4395e, etc.

ESO	CRIRES+ EDPS-GUI tutorial	Doc:	1 rev. 20260918
		Issue:	Issue 1
		Date:	Date Sep 18 2026
		Page:	30 of 30

The users can find in those executable scripts names `cmdline.sh`, together with the necessary parameter files `*.rc` and lists of input fits files `*.sof`. To manually start the execution type at the prompt:

```
./cmdline.sh
```

It is advisable to make copies of all files before making changes.

- **Q9) How can I correct CRIRES+ spectra for the telluric absorption?**

Answer: The telluric correction is not part of the CRIRES+ pipeline. The users have two options: to carry out the correction externally with a telluric standard (if available; must be observed close in time and in airmass to the science target) or to run the `Molecfit` tool². For further information how to run `Molecfit` on the CRIRES+ spectra consult with its own User Manual and with the CRIRES+ Pipeline Manual, Appendix C.

²<https://www.eso.org/sci/software/pipelines/>